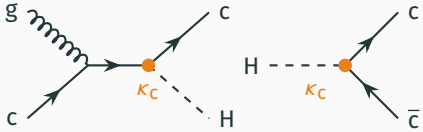
Constraints on Higgs-charm couplings at CMS

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on behalf of the CMS Collaboration
Vrije Universiteit Brussel

Higgs 2026, Louvain-la-Neuve, 28 October 2026



Why the Higgs-charm coupling?

Yukawa couplings today

Since 2012: couplings to t, b, τ measured; evidence for $H \to \mu\mu$ → backup; all SM-like

Why charm?

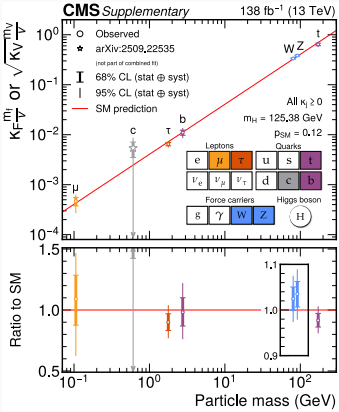
2nd-generation quark Yukawas are the open frontier; charm is the most accessible

- $\mathcal{B}(H \to c\bar{c}) \approx 2.9\%$, about 20× below $H \to b\bar{b}$
- huge QCD backgrounds; c jets are hard to tell from b and light jets (slide 4)

$$\kappa_c = y_c / y_c^{SM} \quad \text{(SM: } \kappa_c = 1\text{)}$$

Flavour-compatible BSM (e.g. spontaneous flavour violation) allows $|\kappa_c| \sim 3$:
current sensitivity is entering this region

→ *Does the Higgs give charm its mass?*



CMS, Rep. Prog. Phys. 89 (2026) 087801 (suppl.); κ_c point: separate $t\bar{t}H$+VH result, not in the fit

How CMS probes κ_c

Direct $H \to c\bar{c}$ (rate $\propto \kappa_c^2/\Gamma_H$) slides 5–9

VH($H \to c\bar{c}$)	PRL 131 (2023) 061801
$t\bar{t}$H($H \to c\bar{c}$) + VH comb.	PRL 136 (2026) 011801
Boosted $H \to c\bar{c}$ (mostly ggH)	PRL 131 (2023) 041801

$t\bar{t}$H+VH: $|\kappa_c| < 3.5$ (2.7)

H + charm production ($cg \to cH$) slides 10–12

cH($H \to \gamma\gamma$)	JHEP 11 (2025) 060
cH($H \to WW$) + comb.	JHEP 08 (2026) 051

κ_c in the production vertex, any H decay

comb.: $|\kappa_c| < 47$ (51)

Indirect: H rates and width slide 13

On-shell $H \to 4\ell$	PRD 112 (2025) 112001
Off-shell + on-shell $H \to 4\ell$ **NEW**	arXiv:2607.23352

$\kappa_c \in [-3.3, 2.8]$ **to** $[-8.6, 7.5]$ (by assumptions)

Rare exclusive decays slide 13

$H \to J/\psi\gamma$, $\psi(2S)\gamma$	PLB 865 (2025) 139462

sensitive to the *sign* of κ_c

$-166 < \kappa_c < 208$

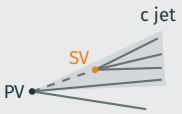
$\mu = \sigma\mathcal{B}/(\sigma\mathcal{B})_{SM}$, Γ_H = total Higgs width; μ limits and κ_c intervals at 95% CL: obs. (exp.)

→ *All κ_c results: Run 2 (13 TeV, up to 138 fb^{-1}); Run 3 so far: better triggers (slide 15)*

Charm tagging: telling c jets from b and light jets

Charm signature

c-hadron mass and lifetime lie **between light and b**:
- secondary vertex (SV), like b jets
- but closer to the primary vertex (PV): easily confused with light jets



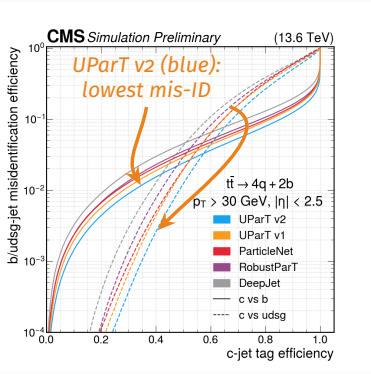
Algorithm evolution at CMS

Likelihood / BDT
JP: track impact parameters
CSV: SV + track variables

Deep NN
DeepCSV: b/c/light classes
DeepJet: all constituents

Particle cloud
ParticleNet: graph network
UParT: transformer

Run 1 → Run 3: richer architectures, inputs closer to raw particles



$t\bar{t}$ simulation, 13.6 TeV; solid: c vs b, dashed: c vs light

Used in this talk: DeepJet (VH resolved, cH), ParticleNet (VH merged, $t\bar{t}$H), DeepDoubleX (boosted H)

Direct $H \to c\bar{c}$: VH($H \to c\bar{c}$)

PRL 131 (2023) 061801, CMS-HIG-21-008

Cleanest signature: $W/Z \to 0, 1, 2$ leptons
V+jets and $t\bar{t}$ normalised in control regions

0L: $Z \to \nu\nu$ | **1L: $W \to \ell\nu$** | **2L: $Z \to \ell\ell$**

below $p_T^H = 300$ **GeV** *above*

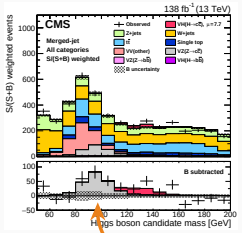
Resolved: two jets

>95% of signal
two DeepJet c-tagged jets
fit to a BDT discriminant

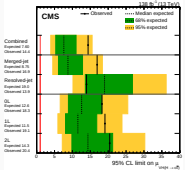
Merged: one large-R jet

<5% of signal, but **less background**
ParticleNet $c\bar{c}$ tag (3 purities) and mass regression; fit to $m(H_{cand})$

$\mu < 14$ (7.6) $\quad 1.1 < |\kappa_c| < 5.5$ ($|\kappa_c| < 3.4$)
95% CL, observed (expected) → backup



VZ($Z \to c\bar{c}$) at 5.7σ validates the method



Most sensitive single channel (exp. $\mu < 7.6$); statistics-limited

Direct $H \to c\bar{c}$: boosted H (ggF-enriched)

PRL 131 (2023) 041801, CMS-HIG-21-012

Boosted regime ($p_T^H > 450$ GeV)

- High-p_T Higgs bosons to suppress QCD; signal 55% ggF, 25% VBF, 20% VH
- $c\bar{c}$ in one large-radius jet; soft drop removes soft, wide-angle radiation and sharpens the jet mass m_{SD}

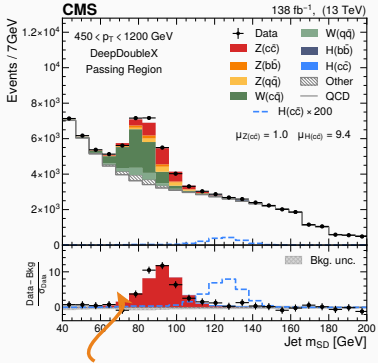
DeepDoubleX tagger

- Deep neural network on low-level jet features: passing (signal-rich) and failing (background-rich) regions
- Decorrelated from m_{SD}: QCD background estimated from data in the failing region

Statistical analysis

- Simultaneous binned fit of m_{SD} in all tagger regions; limit from the profile likelihood ratio

$\sigma\mathcal{B}/(\sigma\mathcal{B})_{SM} < 47\ (39)$ 95% CL, obs. (exp.)
no κ_c interpretation in the paper



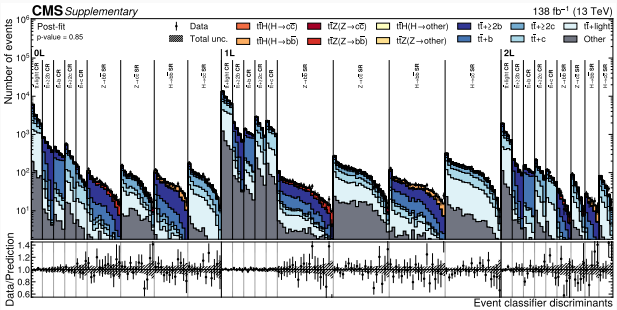
$Z \to c\bar{c}$ observed for the first time in Z+jets, well over 5σ

Passing region; $H \to c\bar{c}$ dashed (×200)

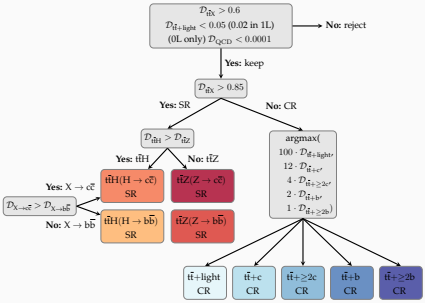
$t\bar{t}H(H \to c\bar{c})$: the whole analysis in one plot

PRL 136 (2026) 011801, CMS-HIG-24-018

3 channels (0L, 1L, 2L) × {5 $t\bar{t}$+jets CRs + 4 SRs}, fitted simultaneously



Post-fit ParT discriminants in all CRs and SRs (p-value 0.85)



Categorisation on $D_{t\bar{t}X}$ (summed signal score):
> 0.85 to an SR by $t\bar{t}H/t\bar{t}Z$and $c\bar{c}/b\bar{b}$; 0.6–0.85 to the CR of the leading $t\bar{t}$+jets score

→ *One fit:* $t\bar{t}H(c\bar{c})$, $t\bar{t}H(b\bar{b})$, $t\bar{t}Z(c\bar{c})$, $t\bar{t}Z(b\bar{b})$ *and all* $t\bar{t}$*+jets components*

$t\bar{t}H(H \to c\bar{c})$: strategy and results

PRL 136 (2026) 011801, CMS-HIG-24-018

Strategy

- Simultaneous $t\bar{t}H(c\bar{c})$ and $t\bar{t}H(b\bar{b})$: same signature apart from the c/b-jet multiplicity; $t\bar{t}Z(c\bar{c})$, $t\bar{t}Z(b\bar{b})$ as standard candles in their own regions
- Every jet in one of 11 ParticleNet tagging categories (Bo–B4, Co–C4, Lo); ParT event classifier with 4 signal and 5 $t\bar{t}$+jets classes → backup
- Leading background $t\bar{t}$+jets, split by extra-jet flavour, each part normalised in its own CR; QCD negligible even in oL (removed by a classifier cut to <1% of $t\bar{t}$+jets) → backup

Regions (on $D_{t\bar{t}X}$, the summed signal score)

- Analysis region $D_{t\bar{t}X} > 0.6$: 4 SRs (> 0.85, by $t\bar{t}H/t\bar{t}Z$and $c\bar{c}/b\bar{b}$) and 5 CRs (0.6–0.85)
- Validation region 0.4–0.6 with the same SR/CR structure: good agreement → backup

$t\bar{t}H(H \to c\bar{c})$, 95% CL obs. (exp.)

$\mu < 7.8$ (8.7)

$|\kappa_c| < 3.0$ (3.3) ($\kappa_b = 1$)

$|\kappa_c| < 4.2$ obs. (κ_b fitted) → backup

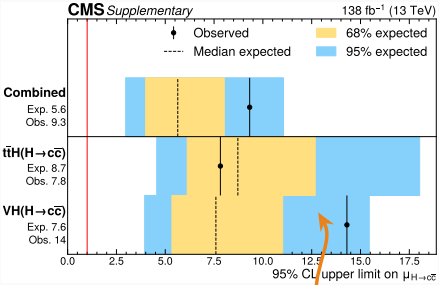
	best fit $\hat{\mu}$	obs. (exp.) signif.
$t\bar{t}H(b\bar{b})$	$0.91^{+0.26}_{-0.22}$	4.4σ (4.5σ)
$t\bar{t}H(c\bar{c})$	-1.6 ± 4.5	–
$t\bar{t}Z(b\bar{b})$	$1.47^{+0.45}_{-0.41}$	3.5σ (2.4σ)
$t\bar{t}Z(c\bar{c})$	$1.02^{+0.79}_{-0.84}$	1.2σ (1.3σ)

Limits per channel and event yields → backup

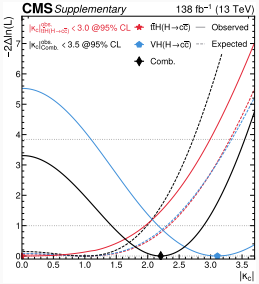
→ *A second direct channel, with expected sensitivity comparable to VH*

$t\bar{t}H$ + VH combination

PRL 136 (2026) 011801, CMS-HIG-24-018



VH fluctuated up ($\hat{\mu}$ = 7.7)



$-2\Delta \ln L$ vs. $|\kappa_c|$: $t\bar{t}H$, VH, comb.; solid obs., dashed exp. (κ_b = 1)

$t\bar{t}H$ + VH, 95% CL

obs. (exp.)

$\mu_{H\to c\bar{c}} < 9.3$ (5.6)

$|\kappa_c| < 3.5$ (2.7)

SM production, $\kappa_b = 1$

$\mu_{H\to c\bar{c}}$ limits: $t\bar{t}H$, VH, combined (inner band 68%)

- Expected μ limit 26% better than VH alone (36% better than $t\bar{t}H$ alone)
- $t\bar{t}H$ alone (3.0) beats the combination (3.5) on observed because VH fluctuated up

→ *Best expected direct sensitivity: $|\kappa_c| < 2.7$, below the BSM-motivated $|\kappa_c| \sim 3$*

H + charm production: κ_c in production

Idea

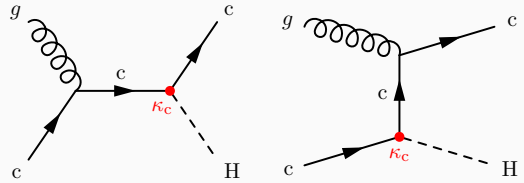
κ_c enters the **production** vertex, independent of the H $\to c\bar{c}$ decay: clean H decay + a c-tagged jet

$\sigma_{cH} = 90\,\text{fb}$: $\sigma\mathcal{B} \approx 0.2\,\text{fb}$ ($\gamma\gamma$), $\approx 20\,\text{fb}$ (WW)

Challenges

- κ_c**-independent H+c** from g$\to c\bar{c}$ splitting in ggH (same final state)
- soft, often forward c jets
- flavour-scheme modelling ($\to$ T. Bevilacqua, next talk)

"Flat direction": κ_c is varied while all other Higgs rates stay at their SM values, so only cH production changes



LO diagrams for cH production; κ_c enters at the marked vertices

cH($H \to \gamma\gamma$)

Analysis

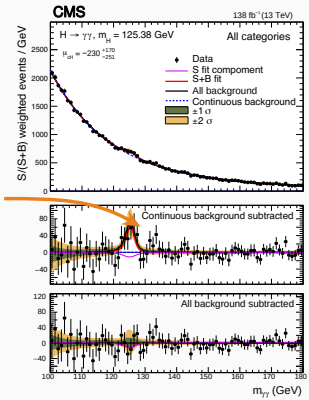
- First search for cH with $H \to \gamma\gamma$, 138 fb^{-1}; DeepJet c tagging
- Two BDTs, trained without c-tag inputs: cH vs. ggH (largest resonant background), cH vs. continuum $\gamma\gamma$+jets
- 9 categories per year (27 in total); signal from fits to $m_{\gamma\gamma}$

$$\mu_{cH} < 243\ (355), \quad |\kappa_c| < 38\ (73)$$

95% CL, observed (expected)

→ *$|\kappa_c| \lesssim 40$, not ~3: $\sigma\mathcal{B} \approx 0.2\,fb$, and ggH H+c mimics the signal*

125 GeV peak: other Higgs processes (mostly ggH); $\hat{\mu}_{cH} = -230$



$m_{\gamma\gamma}$, all categories, weighted by S/(S+B)

cH(H→WW) and the $\gamma\gamma$ + WW combination

JHEP 08 (2026) 051, CMS-HIG-24-009

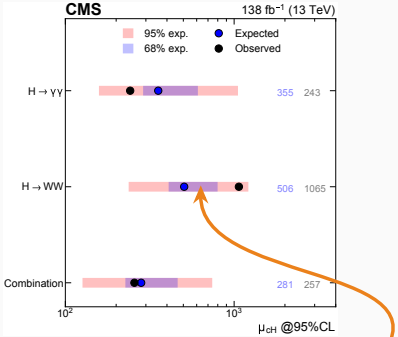
cH(H→WW)

- H→WW→e$\nu\mu\nu$ + c jet (DeepJet); signal regions with 1 or >1 c jet; two BDTs (vs. non-H and vs. other H processes)
- κ_c-independent H+c background fixed to SM: μ_{cH} is the only parameter of interest (1 POI); floating it → backup
- WW alone: $\mu_{cH} < 1065\ (506)$, $|\kappa_c| < 211\ (95)$; post-fit → backup

$\gamma\gamma$ + WW (1 POI), 95% CL obs. (exp.)

$\mu_{cH} < 257\ (281), \quad |\kappa_c| < 47\ (51)$

→ *κ_c from production alone: statistics-limited, independent of the H → c$\bar{c}$ decay*



obs. weaker than $\gamma\gamma$ alone (3σ): WW fluctuated up ($\hat{\mu}_{cH}$ = 390 vs. −230 in $\gamma\gamma$)

μ_{cH} limits: $\gamma\gamma$, WW, combined

Indirect constraints on κ_c

PRD 112 (2025) 112001; arXiv:2607.23352; PLB 865 (2025) 139462

H→ZZ→ 4ℓ: on-shell vs. off-shell

$$\sigma_{\text{on}} \propto \frac{g_p^2\, g_d^2}{\Gamma_H}, \qquad \sigma_{\text{off}} \propto g_p^2\, g_d^2$$

g_p, g_d: production and decay couplings ($g_d \propto \kappa_Z$). Large κ_c adds $c\bar{c} \to H$ and raises Γ_H; on-shell alone, a larger κ_Z can hide it, so $|\kappa_Z| \leq 1$ must be assumed. **Off-shell has no** Γ_H: it fixes κ_Z and removes that assumption

Results, κ_c at 95% CL obs. (exp.)

- **On-shell** ($|\kappa_Z = \kappa_W| \leq 1$): $[-3.7, 3.2]$ ($[-3.8, 3.3]$)
- **NEW On- + off-shell,** $|\kappa_V| \leq 1$: $[-3.3, 2.8]$ ($[-3.5, 2.9]$)
  κ_Z, κ_W free: $[-8.6, 7.5]$ ($[-9.6, 8.3]$)
- H→J/$\psi\gamma$: $-166 < \kappa_c < 208$ (J. Bierkens, Thu); width: F. Errico (Tue); p_T^H → backup

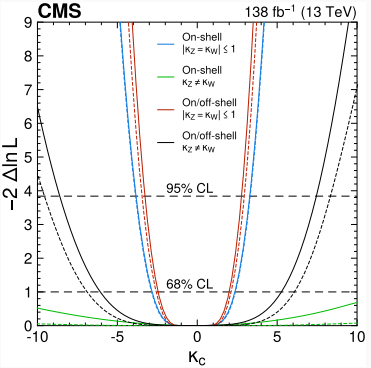


Fig. 8: on-shell (blue, green) and on+off-shell (red, black), with $|\kappa_Z = \kappa_W| \leq 1$ or $\kappa_Z \neq \kappa_W$; solid obs., dashed exp. Text: Table 3, no custodial symmetry

→ *With $|\kappa_V| \leq 1$, H→ 4ℓ matches the direct searches, under different assumptions*

Where we stand: CMS constraints on κ_c

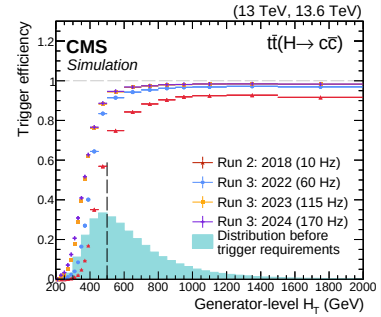
Probe	Reference	κ_c at 95% CL: obs. (exp.)	Main assumptions
$t\bar{t}H$ + VH($H \to c\bar{c}$)	PRL 136 (2026) 011801	$\lvert\kappa_c\rvert < 3.5$ (2.7)	$\kappa_b = 1$, SM production, no BSM decays
$t\bar{t}H$($H \to c\bar{c}$)	PRL 136 (2026) 011801	$\lvert\kappa_c\rvert < 3.0$ (3.3)	same
VH($H \to c\bar{c}$)	PRL 131 (2023) 061801	$1.1 < \lvert\kappa_c\rvert < 5.5$ (3.4)	same
On+off-shell $H \to 4\ell$	arXiv:2607.23352	$[-3.3, 2.8]$ ($[-3.5, 2.9]$)	$\lvert\kappa_V\rvert \leq 1$, $\kappa_{b,t} = 1$, other light κ_q profiled, $\Gamma_{BSM} \geq 0$
On-shell $H \to 4\ell$	PRD 112 (2025) 112001	$[-3.7, 3.2]$ ($[-3.8, 3.3]$)	$\lvert\kappa_Z = \kappa_W\rvert \leq 1$, otherwise same
On+off-shell $H \to 4\ell$	arXiv:2607.23352	$[-8.6, 7.5]$ ($[-9.6, 8.3]$)	κ_Z, κ_W free, otherwise same
cH ($\gamma\gamma$ + WW)	JHEP 08 (2026) 051	$\lvert\kappa_c\rvert < 47$ (51)	flat direction, from μ_{cH} limit
$H \to J/\psi\gamma$	PLB 865 (2025) 139462	$[-166, 208]$ ($[-128, 169]$)	$\kappa_\gamma = 1$, 123 fb^{-1}

The rows use different coupling models and are not directly comparable. $t\bar{t}H$ alone beats the combination on observed because VH fluctuated up; the combination has the best *expected* sensitivity (2.7).

→ *Direct searches and off-shell + on-shell $H \to 4\ell$ both reach $\lvert\kappa_c\rvert \sim 3$, where flavour-compatible BSM scenarios live*

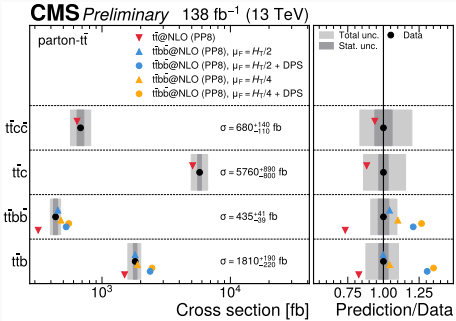
Toward Run 3: better triggers, measured backgrounds

Run 3: ML b/c tagging at the HLT



arXiv:2608.18331, simulation: $t\bar{t}H(H \rightarrow c\bar{c})$ trigger efficiency, Run 2 (2018) vs. Run 3 (2022–24), 13.6 TeV. HLT taggers: J. Daniel's talk (Mon) → backup

Run 2: $t\bar{t}H(H \rightarrow c\bar{c})$ background measured



CMS-PAS-TOP-26-003 **NEW**, Run 2 data (138 fb^{-1}):
$\sigma_{t\bar{t}c\bar{c}} = 680 \pm 40\,(\text{stat})\,^{+130}_{-100}\,(\text{syst})$ fb

→ *Run 3 starts with higher trigger efficiency and a $t\bar{t}$+$c\bar{c}$ background measured in Run 2*

Run 3 and HL-LHC prospects

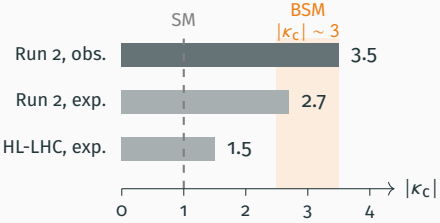
Run 3 (13.6 TeV)

- Direct searches are statistics-limited (VH: 85% of the uncertainty): more data helps directly
- Taggers: UParT rejects more b and light jets than DeepJet and ParticleNet at the same c efficiency (slide 4)
- HLT b/c tagging: higher trigger efficiency, e.g. for $t\bar{t}H(H \to c\bar{c})$ (slide 15)
- $t\bar{t}+c\bar{c}$ background measured in Run 2 data (slide 15)
- ATLAS VBF $H \to c\bar{c}$ already uses $51.5\,fb^{-1}$ of Run 3 data

HL-LHC ($3\,ab^{-1}$ per experiment)

- ATLAS + CMS European Strategy input (arXiv:2504.00672, CMS-HIG-25-002): $VH(H \to c\bar{c})$ $\approx 1.6\sigma$ expected
- **Expected** $|\kappa_c| < 1.5$ (95% CL, 14 TeV): $VH(H \to c\bar{c})$ with improved c tagging (ATLAS study, assumed for both); 30% better than the 2019 Yellow Report

Direct searches: allowed $|\kappa_c|$ at 95% CL



Run 2: $t\bar{t}H$+VH (PRL 136 (2026) 011801); HL-LHC: ATLAS+CMS, $3\,ab^{-1}$ each

Every bar still includes $\kappa_c = 0$

κ_c at $O(1)$: first sensitivity to whether the Higgs gives charm its mass

Summary

The charm Yukawa is the most accessible 2nd-generation quark coupling; CMS probes it in decay, in production and indirectly

Direct $t\bar{t}H(H \to c\bar{c})$ + $VH(H \to c\bar{c})$: $|\kappa_c| < 3.5$ (2.7), the most stringent direct constraint and the best expected sensitivity

Production H+c: $|\kappa_c| < 47$ (51), independent of the $H \to c\bar{c}$ decay

Indirect Off-shell + on-shell $H \to 4\ell$: $\kappa_c \in [-3.3, 2.8]$ with $|\kappa_V| \leq 1$ (about ± 8 without)

Next All κ_c results are Run 2; Run 3 starts with better taggers, higher trigger efficiency and a measured $t\bar{t}$+$c\bar{c}$ background

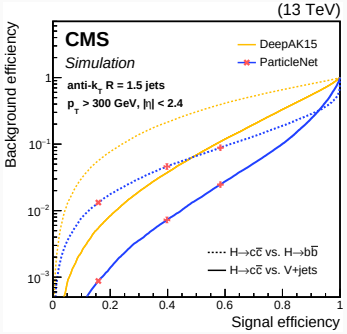
Does the Higgs give charm its mass? Today $|\kappa_c| \lesssim 3$–3.5, entering BSM-motivated territory; HL-LHC expected $|\kappa_c| < 1.5$: approaching, not yet reaching, sensitivity to $\kappa_c = 0$

Thank you!

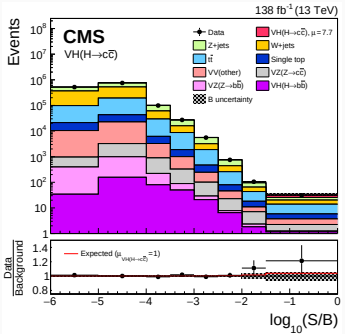
Backup

VH(H → c$\bar{c}$): event categories and sensitivity

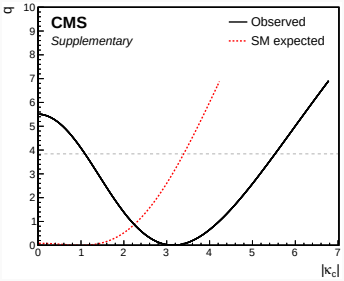
PRL 131 (2023) 061801



ParticleNet merged-jet H → c$\bar{c}$ tagging



Events vs. $\log_{10}(S/B)$



$|\kappa_c|$ scan (supplementary)

Best fit $\mu = 7.7^{+3.8}_{-3.5}$; merged $\mu = 8.7^{+4.6}_{-4.0}$, resolved $\mu = -9.5 \pm 9.6$. $\sigma\mathcal{B} < 0.94$ (0.50) pb. μ limits with CLs; κ_c intervals from profile-likelihood scans.

→ *The upward fluctuation comes from the merged-jet category*

VH($H \to c\bar{c}$): analysis details

PRL 131 (2023) 061801; strategy as in JHEP 03 (2020) 131

Topologies Two, orthogonal: H candidate from a large-R ($R = 1.5$) jet with p_T above (merged) or below (resolved) 300 GeV; same lepton channels

Resolved DeepJet CvsL/CvsB; the two highest-CvsL jets form the H candidate; leading-jet cuts at ≈40% c efficiency (≈4% light, ≈16% b mistag). c-jet energy regression, FSR-jet recovery and a 2L kinematic fit improve the $m(\mathrm{H_{cand}})$ resolution by up to 20%. BDT per channel, fit to the BDT score

Merged ParticleNet $c\bar{c}$ tagger, 4–7× better rejection than DeepAK15; 3 working points (≈58, 40, 16% $c\bar{c}$ efficiency) define 3 purity categories. ParticleNet mass regression: ≈50% better mass resolution. BDTs decorrelated from $m(\mathrm{H_{cand}})$; fit to $m(\mathrm{H_{cand}})$

Backgrounds V+jets and $t\bar{t}$ normalised in dedicated control regions, following the 2016 analysis (V+jets split by jet flavour, free rate parameters)

VZ validation Same analysis, resolved BDTs retrained with VZ as signal, merged unchanged. $\mu_{VZ} = 1.01^{+0.23}_{-0.21}$; 4.4σ merged, 3.1σ resolved, 5.7σ combined (exp. 4.7, 3.3, 5.9)

Uncertainty breakdown

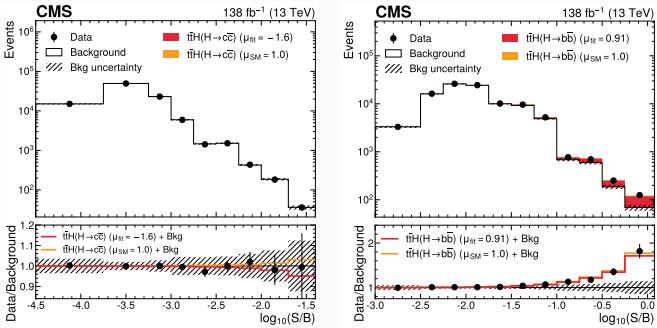
Source	$\Delta\mu/\Delta\mu_{\mathrm{tot}}$
Statistical	**85%**
Bkg. normalisations	37%
Experimental	48%
Simulated sample sizes	37%
c-jet identification	23%
Jet energy scale, resolution	15%
Simulation modelling	11%
Luminosity	6%
Lepton identification	4%
Theory	22%
Backgrounds	17%
Signal	15%

For $\hat{\mu} = 7.7^{+3.8}_{-3.5}$; simulation size driven by negative weights in NLO V+jets

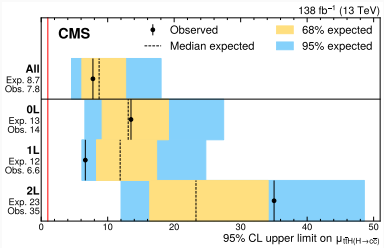
→ *Statistics-limited; the c-tagging and simulation-size systematics come next*

$t\bar{t}H(H \to c\bar{c})$: event yields

PRL 136 (2026) 011801



Event yields vs. $\log_{10}(S/B)$; the $t\bar{t}H(c\bar{c})$ panel shows the published best fit $\hat{\mu} = -1.6 \pm 4.5$



$\mu_{t\bar{t}H(c\bar{c})}$ limits per channel (inner band 68%)

→ $t\bar{t}H(c\bar{c})$ *best fit* $\hat{\mu} = -1.6 \pm 4.5$*: compatible with zero and with the SM*

$t\bar{t}H(H \to c\bar{c})$: flavour tagging and by-products

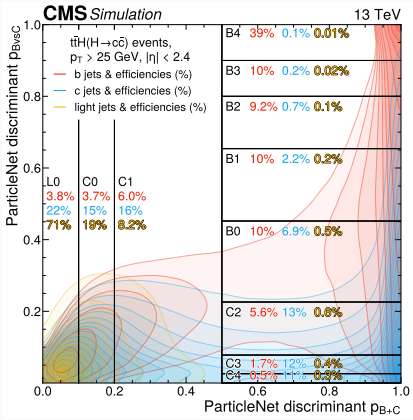
PRL 136 (2026) 011801

Tagging

- Each jet gets ParticleNet scores; the (b+c vs. light, b vs. c) plane is split into 2D categories
- ParT event classifier separates $t\bar{t}H(c\bar{c})$, $t\bar{t}H(b\bar{b})$, $t\bar{t}Z$, $t\bar{t}$+HF

By-products

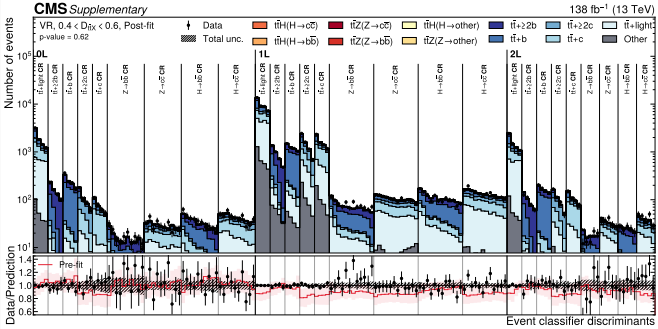
- $t\bar{t}Z(Z \to c\bar{c})$ as a standard candle: $\mu = 1.0 \pm 0.8$
- $t\bar{t}H(H \to b\bar{b})$ $\mu = 0.91^{+0.26}_{-0.22}$, 4.4σ (4.5 exp.)



b, c, light jets in the ParticleNet (b+c vs. light, b vs. c) plane and the 2D tagging categories

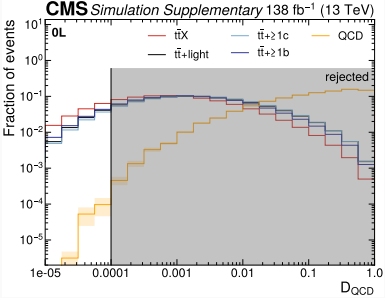
$t\bar{t}H(H \to c\bar{c})$: validation region and QCD rejection

PRL 136 (2026) 011801 (supplementary)



Validation region $0.4 < D_{t\bar{t}X} < 0.6$, same SR/CR structure (CR-like 0.40–0.58, SR-like 0.58–0.60): good agreement

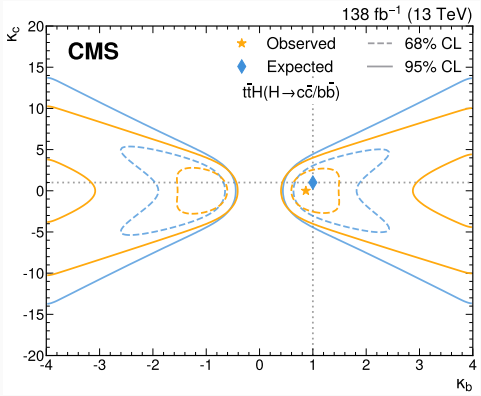
- c-enriched regions need ≥2 c jets ($t\bar{t}$+c: ≥1); CR choice weights 100, 12, 4, 2, 1 on the $t\bar{t}$+light, +c, +≥2c, +b, +≥2b scores
- $t\bar{t}$+b, +≥2b from a dedicated 4FS $t\bar{t}b\bar{b}$ sample



0L only: $D_{QCD} < 10^{-4}$ (grey area rejected) suppresses QCD by nearly four orders of magnitude, to <1% of $t\bar{t}$+jets; QCD simulation is used only to train the classifier

$t\bar{t}H(H \to c\bar{c}/H \to b\bar{b})$: (κ_b, κ_c) plane

PRL 136 (2026) 011801



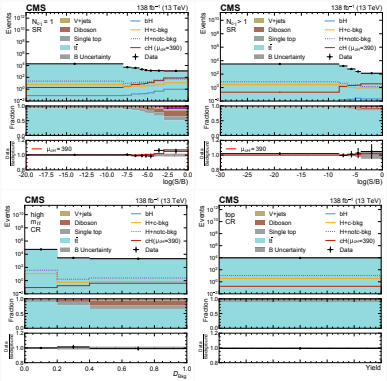
orange: observed, blue: expected; dashed 68%, solid 95% CL

- Simultaneous $t\bar{t}H(H \to b\bar{b})$ + $t\bar{t}H(H \to c\bar{c})$ fit constrains both Yukawas
- Signal strengths: $t\bar{t}H(b\bar{b})$ $0.91^{+0.26}_{-0.22}$; $t\bar{t}H(c\bar{c})$ -1.6 ± 4.5; $t\bar{t}Z(b\bar{b})$ $1.47^{+0.45}_{-0.41}$; $t\bar{t}Z(c\bar{c})$ $1.02^{+0.79}_{-0.84}$

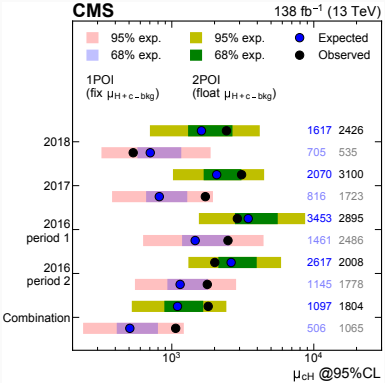
$|\kappa_c| < 4.2$ obs. with κ_b fitted

cH(H→WW): post-fit distributions and per-period limits

JHEP 08 (2026) 051



1-POI post-fit distributions of the BDT discriminants

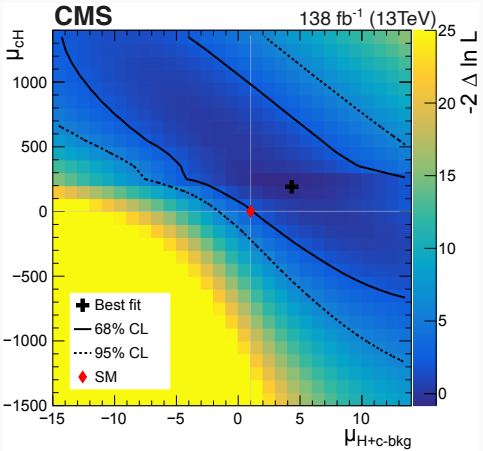


μ_{cH} limits per data-taking period, 1 and 2 POI

→ *Best fit (1 POI):* $\mu_{cH} = 390^{+380}_{-270}$, *within 1.5σ of the SM; statistics-limited*

cH(H→WW): floating the H+c background

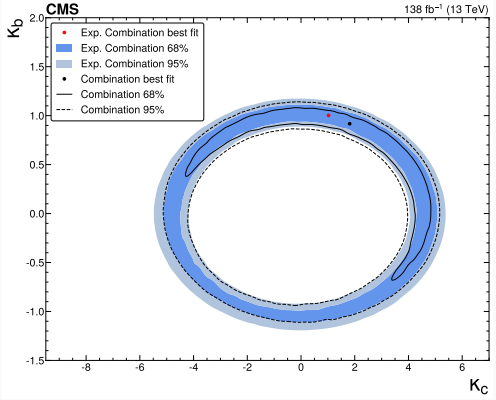
- **1 POI** only μ_{cH} floats; the κ_c-independent H+c background (from ggH etc.) is fixed to its SM prediction
- **2 POI** H+c background also floats; correlation with cH: −87%
- 2-POI limit: $\mu_{cH} < 1804$ (1097)
- The combination with $\gamma\gamma$ uses the 1-POI fit



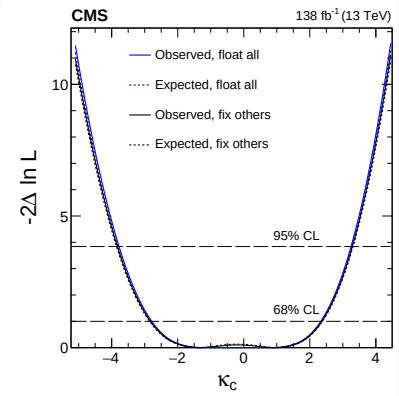
WW: μ_{cH} vs. $\mu_{H+c\ bkg}$

→ *Floating the H+c background weakens the WW limit from 1065 to 1804*

Indirect: Higgs p_T spectrum and on-shell H$\to 4\ell$



JHEP 04 (2026) 093: (κ_b, κ_c) from the Run 2 p_T^H combination ($\gamma\gamma$, ZZ, resolved $\tau\tau$), $\mathcal{B}$ coupling-dependent; 2D contours only



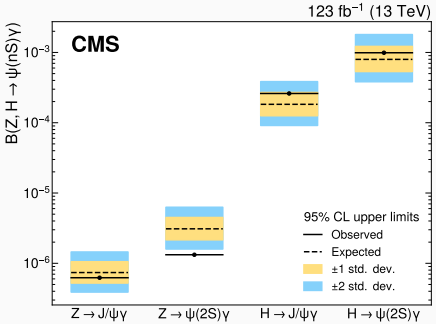
PRD 112 (2025) 112001 (γH + on-shell H$\to 4\ell$): κ_c scan; solid obs., dashed exp.

Rare exclusive decays: H→J/$\psi\gamma$, ψ(2S)γ

PLB 865 (2025) 139462, CMS-SMP-22-012

- Direct ($\propto \kappa_c$) and indirect (H→ $\gamma\gamma^* \to \gamma$ J/ψ) amplitudes interfere ⇒ sign sensitivity
- 123 fb^{-1}: $\mathcal{B}(H \to J/\psi\gamma) < 2.6 \times 10^{-4}$, $\mathcal{B}(H \to \psi(2S)\gamma) < 9.9 \times 10^{-4}$

$\kappa_c/\kappa_\gamma \in (-157, 199)$; with $\kappa_\gamma = 1$:
$\kappa_c \in (-166, 208)$
exp. $(-128, 169)$, 95% CL



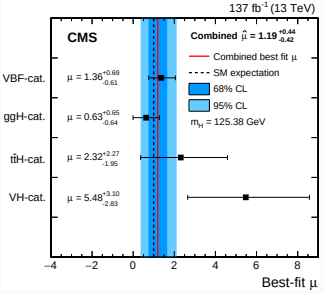
95% CL upper limits on $\mathcal{B}$(Z, H→ $\psi(n\mathrm{S})\gamma$), obs. and exp.

H→ $\mu\mu$: evidence for a 2nd-generation Yukawa

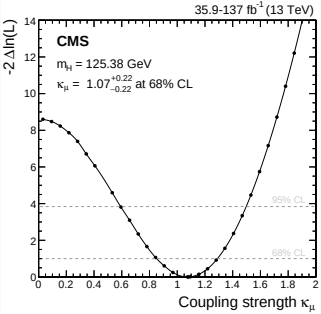
JHEP 01 (2021) 148, CMS-HIG-19-006

- 137 fb^{-1}, four categories: ggH, VBF, VH, $t\bar{t}H$
- Excess of **3.0**σ (2.5σ exp.): first evidence for H decays to 2nd-generation fermions
- $\mu = 1.19^{+0.40}_{-0.39}\,(\text{stat})^{+0.15}_{-0.14}\,(\text{syst})$
- $\kappa_\mu = 1.07$; 95% CL: $0.59 < \kappa_\mu < 1.50$ (6 free κ, SM total width)

Rare decays incl. $\mu\mu$: L. D'Eramo's plenary (Wed)



$\hat{\mu}$ per production category and combined

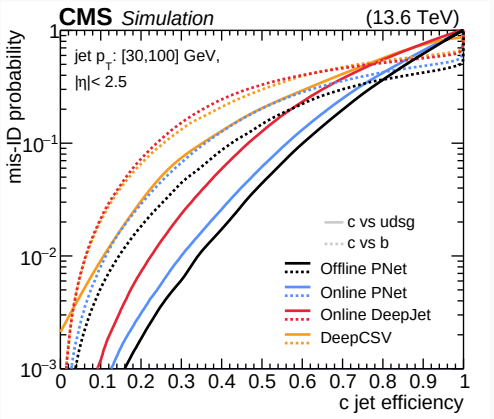


$-2\Delta \ln L$ vs. κ_μ

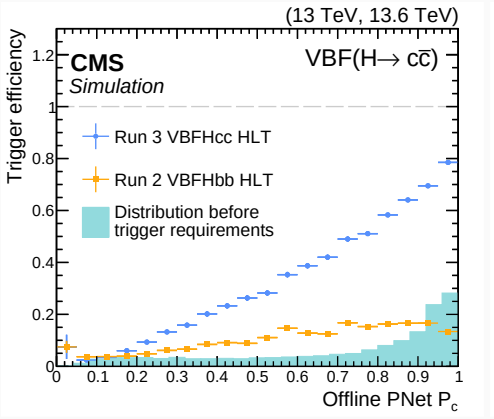
→ *The muon Yukawa is SM-like within about 20% (68% CL); charm is the next 2nd-generation test*

Run 3 HLT flavour tagging

arXiv:2608.18331, CMS-BTV-25-002



Simulation, jet p_T 30–100 GeV: light (solid) and b (dashed) misidentification vs. c-jet efficiency, online vs. offline taggers



Simulation: VBF H→ $c\bar{c}$ trigger efficiency vs. offline ParticleNet c score, Run 3 VBF H→ $c\bar{c}$ vs. Run 2 VBF H→ $b\bar{b}$ trigger

Likely questions

- **How large is the VH excess?** Best fit $\mu_{VH} = 7.7^{+3.8}_{-3.5}$, about 1.9σ above $\mu = 1$ (estimated from the quoted uncertainty; the paper calls it a small upward fluctuation). This is why VH alone excludes $|\kappa_c| < 1.1$ and why $t\bar{t}H$ alone (3.0) beats the combination (3.5) on observed
- **What limits the direct searches?** Statistics first. $t\bar{t}H(c\bar{c})$: statistical 74% of the total uncertainty, then $t\bar{t}$+c and $t\bar{t}$+light modelling (≈32%, ≈29%). VH($c\bar{c}$): simulation sample size (≈37%), c-tagging efficiency (≈23%), theory (≈22%)
- **And ATLAS?** VBF + VH($H \to c\bar{c}$) (arXiv:2511.21911): $|\kappa_c| < 4.7$ (3.9 exp.), same assumptions as CMS (other couplings SM, no BSM decays); its VBF part includes 51.5 fb^{-1} of Run 3 data
- **Why is cH an order of magnitude weaker than** $H \to c\bar{c}$**?** cH $\sigma\mathcal{B}$ is tiny (≈0.2 fb in $\gamma\gamma$) and ggH produces the same H+c final state independently of κ_c; in $H \to c\bar{c}$, κ_c changes the branching fraction of Higgs bosons from much larger production modes (VH, $t\bar{t}H$)
- **Why is cH($\gamma\gamma$) obs. (38) much better than exp. (73)?** The best fit is negative, $\hat{\mu}_{cH} = -230^{+170}_{-251}$
- **Why no** κ_c **from boosted** $H \to c\bar{c}$**?** The paper quotes only μ; the result is ggH-dominated
- **How much of the Run 3 trigger gain is the tagger?** Part of it is rate budget: 10 Hz in 2018 vs. 60, 115, 170 Hz in 2022, 2023, 2024
- **Are the H→** 4ℓ **intervals sign-sensitive?** Not really: the small asymmetry comes from charm-loop interference in ggH and H→gg; the constraint is essentially on κ_c^2

References

- VH($H \to c\bar{c}$): CMS, PRL 131 (2023) 061801, arXiv:2205.05550 (HIG-21-008)
- $t\bar{t}H(H \to c\bar{c})$ + VH combination: CMS, PRL 136 (2026) 011801, arXiv:2509.22535 (HIG-24-018)
- VH($H \to c\bar{c}$) with 2016 data (analysis strategy): CMS, JHEP 03 (2020) 131, arXiv:1912.01662 (HIG-18-031)
- Boosted $H \to c\bar{c}$: CMS, PRL 131 (2023) 041801, arXiv:2211.14181 (HIG-21-012)
- cH($H \to \gamma\gamma$): CMS, JHEP 11 (2025) 060, arXiv:2503.08797 (HIG-23-010)
- cH(H→WW) and combination: CMS, JHEP 08 (2026) 051, arXiv:2508.14988 (HIG-24-009)
- Differential combination Run 2: CMS, JHEP 04 (2026) 093, arXiv:2504.13081 (HIG-23-013)
- γH + $H \to 4\ell$ light Yukawas: CMS, PRD 112 (2025) 112001, arXiv:2502.05665 (HIG-23-011)
- Off-shell + on-shell $H \to 4\ell$: CMS, arXiv:2607.23352, submitted to PRD (HIG-25-003)
- $H \to J/\psi\gamma$: CMS, PLB 865 (2025) 139462, arXiv:2411.15000 (SMP-22-012)
- $H \to \mu\mu$: CMS, JHEP 01 (2021) 148, arXiv:2009.04363 (HIG-19-006)
- Jet flavour tagging in Run 3: CMS-DP-2025-081; DeepJet: E. Bols et al., JINST 15 (2020) P12012
- HLT flavour tagging: CMS, arXiv:2608.18331 (BTV-25-002)
- $t\bar{t}$+b/c cross sections: CMS-PAS-TOP-26-003
- ATLAS VBF + VH($H \to c\bar{c}$): ATLAS, arXiv:2511.21911
- HL-LHC: ATLAS and CMS, arXiv:2504.00672 (HIG-25-002)
- Higgs combination: CMS, Rep. Prog. Phys. 89 (2026) 087801, arXiv:2602.18611 (HIG-21-018)
- Spontaneous flavour violation: arXiv:2410.05236
- HL-LHC Yellow Report: M. Cepeda et al., CERN Yellow Rep. Monogr. 7 (2019) 221, arXiv:1902.00134